

Entanglement Entropy and its Evolution in the Deep Inelastic Scattering process in QCD

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*You look up or down, you see the **PAST**.*

*You look around, you see the **PRESENT**.*

*You know where to look to see the **FUTURE**.*

Acknowledgements

I would like to wholeheartedly thank my advisor, Prof. **Apoorva D. Patel**, for his invaluable support, feedback and patience throughout this project. Given my academic background, I would not have dared to undertake this work if it were not for him. I am honored to have met him, studied under him, and had the opportunity to work under his guidance.

I would like to extend my warm regards and heartfelt thanks to all the professors from whom I have gained knowledge, and to my fellow batchmates who have journeyed with me over the past two years, sharing both academic and personal experiences.

I would like to thank the staff at the CHEP office, particularly Mr. Vinayak and Mr. Saravana, for providing the necessary resources to carry out my work.

It is important to mention that I have little working knowledge of foundational theories such as Relativistic Quantum Mechanics, Quantum Field Theory (QFT), and Quantum Electrodynamics (QED). As a result, this thesis emphasizes qualitative understanding over rigorous mathematical formalism. Given the limited timeframe available for this project, it was not feasible to study all of these subjects in depth. Instead, I focused on learning only the most essential concepts, made reasonable assumptions where necessary, and applied them as required.

Honestly, I feel that choosing this project was nothing short of a radical and risky decision in every sense, but I'm glad I took the shot.

Abstract

Quantum Chromodynamics (QCD), the theory describing strong interactions between quarks and gluons, poses persistent challenges due to its non-abelian nature and confinement property. At high energies, particularly in the regime of Deep Inelastic Scattering (DIS), the partonic structure of hadrons becomes accessible, and parton distribution functions (PDFs) exhibit rich evolution governed by equations like DGLAP and BFKL. These equations predict increasing gluon densities at small Bjorken- x .

This thesis explores the intersection of QCD and Quantum Information Theory (QIT), particularly through the lens of entanglement entropy. Starting from Light-Cone Perturbation Theory (LCPT) and the parton model, we reformulate the hadronic wavefunction as a mixed density matrix from which entanglement entropy can be computed. In the inclusive DIS setting, the inability of high-energy probes to access phase information of Fock components leads to information scrambling, naturally giving rise to von Neumann entropy. We show that this entropy corresponds to the gluon distribution as $S = \ln(xG(x))$, suggesting that gluon multiplicity and partonic entanglement are intimately connected.

Using Mueller's dipole model, a 1+1 dimensional toy model of QCD evolution is studied, where gluon cascades are described by a probabilistic branching process. The resulting entanglement entropy, derived from the dipole multiplicity distribution, grows linearly with rapidity, indicating maximal entanglement at small x . The numerical results show agreement with experimental hadron multiplicity data, reinforcing confidence in the model.

This thesis thus presents a novel viewpoint: at small- x , the proton behaves as a maximally entangled state of partons. The entanglement entropy provides a quantum-information-theoretic observable that can bridge theory and experiment, offering new insights into the inner structure of matter and the dynamics of QCD at high energies.

Organization of the content

Chapter 1 introduces to few mathematical preliminaries required to understand QCD and presents the problem statement. Reference for this chapter is [1] and [2]

Chapter 2 delves into the amateur formalism of QCD at high energy including Deep Inelastic Scattering and introduces evolution equations. Reference for this chapter is [3].

Chapter 3 summarizes the theoretical and experimental aspects of applying quantum information theory (QIT) tools to QCD. Reference for this chapter is [4]

Chapter 4 presents the widely used Mueller's dipole model in QCD as 1+1 toy model along with the solution to the cascade equation in the large rapidity limit.

Chapter 5 generalizes the above 1+1 dimensional model and introduces additional formalism to account for a moving rapidity window. It introduces the Universal and KL entropies and explores the model's ability to explain experimental observations.

Chapter 6 has Conclusions, Future work and Appendix.

Contents

Acknowledgements	i
Abstract	ii
1 Introduction	1
1.1 Particle Physics - History	1
1.2 Standard Model (SM) of Particle Physics	1
1.3 Production and Detection of Particles - High Energy Physics	3
1.3.1 S-matrix	4
1.3.2 Cross-section (σ)	5
1.4 Quantum Chromodynamics (QCD)	5
1.5 Problem statement	6
2 QCD at High Energy	7
2.1 Light Cone Perturbation theory (LCPT)	7
2.2 Deep Inelastic Scattering (DIS)	8
2.2.1 Key Kinematic variables	9
2.2.2 Cross-section and Structure functions	10
2.2.3 Parton model and Bjorken scaling	12
2.3 QCD evolution equations	14
2.3.1 The Dokshitzer–Gribov–Lipatov– Altarelli–Parisi (DGLAP) evolution equation	15
2.3.2 Double logarithmic approximation	19
2.3.3 Insight into other evolution equations	21
3 Quantum Information Theory for DIS - An Overview	23
3.1 Theoretical Framework	23
3.2 Experimental Evaluation	26
4 Modeling QCD Evolution	27
4.1 1+1 Toy model of non-linear QCD evolution	28
4.2 3+1 dimensional Model	31
5 QCD evolution of Entanglement Entropy	33
5.1 Entanglement Entropy in Inclusive DIS	33
5.1.1 Definition of a local dipole probability distribution	35
5.1.2 Definition of local entropy	36
5.2 Experimental data	37
5.3 Numerical Results	39
6 Closing Remarks	42
6.1 Conclusions	42

6.2 Future work	42
Appendix	43
References	44

Chapter 1

Introduction

1.1 Particle Physics - History

The origin of particle physics can, in retrospect, be traced back to the discovery of the electron by J.J. Thomson in 1897, using cathode ray tube experiments. This groundbreaking finding marked the beginning of an era that led to the identification of fundamental (elementary - not known to have any smaller components) particles, eventually giving rise to a distinct branch of physics known as particle physics. Over time, the development of sophisticated theoretical frameworks such as Special Relativity, Relativistic Quantum Mechanics, and Quantum Field Theory (QFT) fundamentally transformed our understanding of these particles. Quantum Field Theory, which is formulated using the Lagrangian or Hamiltonian formalism, introduced the modern viewpoint that particles are excitations or fluctuations of underlying quantum fields. This field-based perspective is now the most widely accepted description of fundamental particles and their interactions, providing the foundation for describing the dynamics and behavior of particles within the framework of the Standard Model of particle physics.

1.2 Standard Model (SM) of Particle Physics

The current framework of Standard Model can be summarized into the following figure which is very ubiquitous,

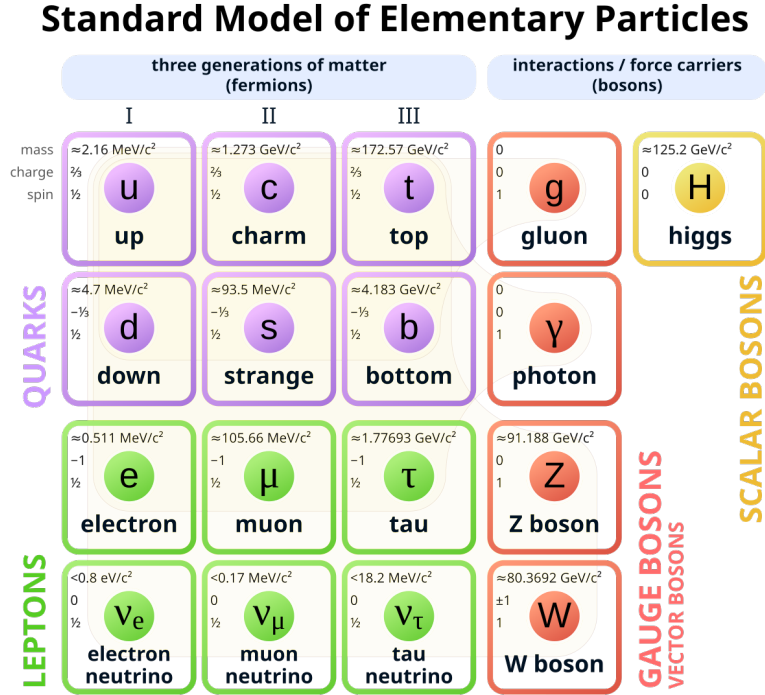


Figure 1.1: Standard Model

All the matter particles turn out to be fermions (half-integer spin particles) and all the force carriers turn out to be bosons (integer spin particles). Quarks, Leptons come in various flavors, primarily organized based on their mass. Quarks and gluons carry color charge and participate in strong interactions, while leptons do not. For every matter particle there is an antiparticle as well (opposite electric charge, color charge, baryon number, lepton number, etc. to their corresponding particles).

The fundamental forces—strong, weak, electromagnetic, and gravitational—are mediated by gauge bosons: gluons for the strong force, W and Z bosons for the weak force, photons for electromagnetism, and gravitons (theoretical) for gravity. The Standard Model also includes the Higgs boson, which is responsible for giving mass to particles through the Higgs mechanism. Together, these particles and interactions form a consistent framework describing nearly all known phenomena in particle physics.

In the language of quantum field theory (QFT), the table below illustrates how different fields give rise to various particles in the Standard Model,

Type of Field	Example Fields	Particles
Spinor Fields	Electron, quark, neutrino fields	Electrons, neutrinos, quarks
Vector Fields	Photon, W, Z, gluon fields	Photon, W^+ , Z^0 , gluons
Scalar Field	Higgs field	Higgs boson

Table 1.1: Basic mapping between field types and particles in the Standard Model.

These fields are solutions of various field equations, for example, Spinor fields (Dirac Spinors) satisfy the Dirac equation etc.

1.3 Production and Detection of Particles - High Energy Physics

Besides cosmic rays and nuclear reactors, particle colliders around the world are the primary sources for producing a wide range of particles—for example, the Large Hadron Collider (LHC) and the Hadron-Electron Ring Accelerator (HERA) etc. A dedicated branch of study, known as collider physics, focuses on the detailed workings of these experimental setups. In what follows, I present some points regarding these experiments and highlight certain key connections between experimental observations and theory, which will be further useful.

In high-energy colliders, known particles are accelerated to velocities close to the speed of light and then collided with one another. The goal is to break them apart, potentially revealing their internal structure or producing new particles, which are then detected using sophisticated detectors.

Let us consider electron-proton scattering, this process can be viewed from a different perspective. Assuming the electron is a "point-like" particle we shift our focus towards proton. In this view, the electron serves as a probe scattering off the proton and potentially revealing its internal structure. The electron emits a photon in this process. The resolution is limited by the Debroglie wavelength of the photon which is

$$\lambda = \frac{h}{q} \tag{1.3.1}$$

where q is the momentum of the photon and h is Planck's constant.

So, the resolution of the microscope is high when the momentum of the photon is large, and it can probe smaller distances.

These type of scattering experiment provides insight into the constituents of the proton and the dynamics that govern their interactions. Although the explanation focuses on electron-proton scattering, it is

generally applicable to any particle scattering process.

The detection of particles involves a range of detectors, from simple devices like bubble chambers, cloud chambers etc to advanced systems such as the Compact Muon Solenoid (CMS), the H1 detector etc.

The following subsection discusses a generic description of 'cross sections,' which are quantities extracted from experimental measurements and its relation with S-matrix,

1.3.1 S-matrix

The S-matrix (Scattering matrix) is a mathematical object in quantum field theory and quantum mechanics that relates the input state and output state of a physical system undergoing a scattering process. The elements of the S-matrix are usually calculated perturbatively. It is necessarily a Unitary matrix.

The S-matrix acts as an operator mapping incoming (asymptotic past) states to outgoing (asymptotic future) states, mathematically (in Heisenberg's picture) the S-matrix transition amplitude is defined as

$$\langle f|S|i\rangle \quad (1.3.2)$$

where $|i\rangle$: input state, $|f\rangle$: final state .

Furthermore, in a free theory, where there are no interactions, the S-matrix is simply the identity matrix $\mathbb{I}$. We can therefore write

$$S = \mathbb{I} + iT, \quad (5.16)$$

where T is called the Transfer/Transition matrix and describes deviations from the free theory.

Since the S-matrix should vanish unless the initial and final states have the same total 4-momentum, it is helpful to factor out an overall momentum-conserving delta function:

$$T = (2\pi)^4 \delta^{(4)}\left(\sum p\right) \mathcal{M}, \quad (5.17)$$

Here, $\delta^{(4)}\left(\sum p\right)$ is shorthand for

$$\delta^{(4)}\left(\sum p_i^\mu - \sum p_f^\mu\right),$$

where p_i^μ are the initial particles' momenta and p_f^μ are the final particles' momenta.

In this way, we can focus on computing the non-trivial part of the S-matrix, $\mathcal{M}$. In quantum field theory, "matrix elements" usually refer to $\langle f|\mathcal{M}|i\rangle$. Thus, we have

$$\langle f|S - \mathbb{I}|i\rangle = i(2\pi)^4 \delta^{(4)}\left(\sum p\right) \langle f|\mathcal{M}|i\rangle. \quad (1.3.3)$$

we define $|\mathcal{M}|^2 \equiv |\langle f|\mathcal{M}|i\rangle|^2$, which is of interest as mentioned above and it is related to the cross section of the experiment, as discussed below.

1.3.2 Cross-section (σ)

The strength of a particular interaction between two particles is specified by the cross section (interaction cross section). The experimental calculation of cross-section depends on number of particles scattered, number density in the beam and velocity of the beam. The relation between cross-section and matrix element is as follows,

$$\frac{d\sigma}{d\Omega} \propto |\mathcal{M}|^2 \quad (1.3.4)$$

The derivation relies on the second form of Fermi's Golden Rule within perturbation theory.

The above description of the S-matrix, cross sections, and their interconnections is sufficient for our purpose, at least at a qualitative level.

1.4 Quantum Chromodynamics (QCD)

As discussed earlier, particle interactions are described within the framework of quantum field theory (QFT), which includes specialized branches such as quantum electrodynamics (QED) for the electromagnetic interaction. Another key branch is quantum chromodynamics (QCD), developed to describe the strong interaction, the force that acts between quarks (spin- $\frac{1}{2}$ fermions) and is mediated by gluons (spin-1 bosons). The QCD Lagrangian encompasses quark and gluon fields along with their interactions.

It is worth highlighting a fundamental difference between quantum electrodynamics (QED) and quantum chromodynamics (QCD): the presence of color charge in QCD. The term 'chromo' in QCD refers to this color charge, which is carried by quarks and gluons. Experimental observations suggest that there are three distinct color charges, leading to significant differences in the underlying mathematical structure. While QED is based on the Abelian gauge group $U(1)$, which governs the dynamics of electrically charged particles, QCD is formulated using the non-Abelian gauge group $SU(3)$, reflecting the richer structure of the strong interaction. It is this non-abelian structure which accounts for the interaction of gluons with themselves unlike in the case of photons (mediators of the electromagnetic force).

There are two main characteristic features in QCD. They are:

- **Confinement:** quarks and gluons do not exist as free particles in nature but combine into color neutral bound states (hadrons) instead.

- **Asymptotic freedom:** quarks and gluons tend to interact more weakly over short distances (high energies) and more strongly over longer distances (low energies).

1.5 Problem statement

The above-mentioned characteristics are experimentally observed but not yet fully understood, making them a subject of study within the scope of QCD.

Earlier work on QCD involved Monte Carlo lattice calculations, a fundamentally non-perturbative approach that provides a crucial tool for studying QCD at low-energy scales where traditional perturbation theory breaks down. This method works by discretizing space-time into a finite lattice. Lattice QCD is entirely numerical and typically yields quantities such as form factors as expectation values and the decomposition of these form factors into parton distribution functions is not feasible. Additionally, computational efforts rapidly increases for simulations involving larger or more complex systems, which is a significant challenge.

A new approach has recently been proposed to employ ideas from Quantum Information Theory to tackle unanswered questions in QCD with the help of models. My work has focused on understanding this line of thought.

Chapter 2

QCD at High Energy

QCD at High Energy deals with the behavior of quarks and gluons in high energy collisions. This chapter introduces topics like Deep Inelastic Scattering, Parton model and few evolution equations of QCD.

Although various types of particle collision are studied in colliders, our focus is on QCD, specifically in the context of lepton-hadron scattering, most commonly electron-proton scattering. This particular scattering will be the point of contention henceforth.

2.1 Light Cone Perturbation theory (LCPT)

Most calculations in high-energy QCD are performed using the rules of Light-Cone Perturbation Theory (LCPT). This approach serves as a departure from the conventional use of Feynman rules in standard quantum field theory calculations. A few key postulates of LCPT are outlined below, as they are useful in getting familiar with subsequent sections and chapters.

- **Light-cone coordinates (Front Form dynamics)[5]** : The instant form coordinates (t, x, y, z) are transformed to (x^+, x, y, x^-) where time t and longitudinal coordinate z (choice: proton moves along z -axis) transform to light-cone time x^+ and light-cone space x^- ,

$$x^+ = \frac{1}{\sqrt{2}}(t + z), \quad x^- = \frac{1}{\sqrt{2}}(t - z) \tag{2.1.1}$$

- **Hamiltonian Formulation:**

– Light-cone Hamiltonian P^- evolves states in light-cone time (x^+) .

- Quantization is done at fixed $x^+ = c$, usually $x^+ = 0$.
- **Fock State Expansion:** Hadrons are expanded in fock space basis. (see chapter 3)
- **Momentum Conservation:** Light-cone momentum is conserved. Included in the form of delta functions.
- **Transverse Momentum Integrals:** Loop integrals in LCPT are done over transverse momenta and light-cone fractions, like $\int \frac{dx d^2 \vec{k}_\perp}{2(2\pi)^3 x}$.
- **Gauge Choice:** The usual gauge choice is $A^+ = 0$ where A^μ refers to gauge field.

There are LCPT rules analogous to the Feynman rules, which yield the same results as standard Feynman diagram calculations, thereby demonstrating their equivalence.

2.2 Deep Inelastic Scattering (DIS)

There is a regime in these collision experiments called Deep Inelastic Scattering (or DIS) which is of our interest. Here "Deep" indicates the probing nature (observing the inside of the proton) and "Inelastic" corresponds to the transfer of energy/momentum to the proton, Whilst elastic regime is where there is no transfer of momentum between particles.

The energy/momentum transfer happens via something called as a virtual photon (γ^*). A virtual photon is a photon that does not follow Einstein's mass-energy equivalence relation ($E \neq |P|$). Quantum Electrodynamics is assumed to be exact and all about electron during this process is known and hence we can separate it. So now the process can be termed as "photon-proton" scattering

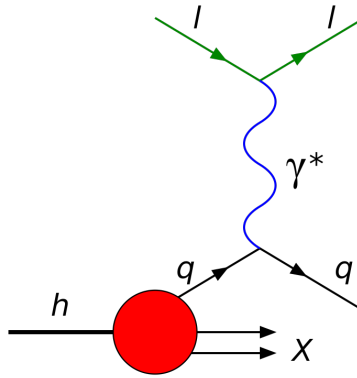


Figure 2.1: Feynman diagram representing DIS

The above reaction is written as follows,

$$e + h \longrightarrow e' + X \quad (2.2.1)$$

e : incoming electron

h : incoming hadron (proton)

e' : outgoing electron

X : formed hadron

2.2.1 Key Kinematic variables

This subsection has the key kinematic variables that are used in calculations, needless to say all are Lorentz covariant independent quantities. Consider the labeled diagram for the DIS,

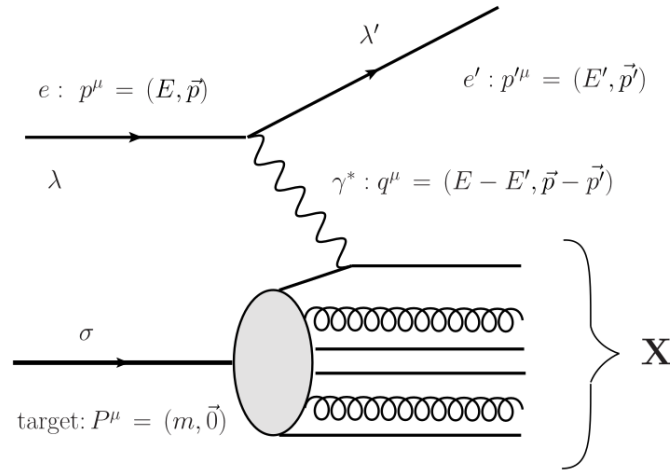


Figure 2.2: Feynman diagram describing deep inelastic electron–proton scattering.

In the above figure 2.2 the momentum labels of the lines correspond to the frame in which the target proton is at rest. The wavy line denotes the virtual photon propagator while the corkscrew lines denote the gluons inside the proton.

The useful Lorentz-invariant quantities are,

$$Q^2 \equiv -q^2, \quad x_{\text{Bj}} \equiv \frac{Q^2}{2P \cdot q} \quad (2.2.2)$$

other Lorentz-invariant (but not independent) quantity can also be defined,

$$\hat{s} \equiv (P + q)^2 = 2P \cdot q + q^2 + m^2, \quad x_{Bj} = \frac{Q^2}{\hat{s} + Q^2 - m^2} \quad (2.2.3)$$

which is the center-of-mass energy squared of the $\gamma^* + p$ reaction.

- **Q^2 :** It is called the virtuality of the photon. It determines the resolution, in the "microscope" description. Also, the theory separates into perturbative QCD ($Q^2 \gg \Lambda^2$) and non-perturbative QCD ($Q^2 \sim \Lambda^2$). Where $\Lambda \sim 200 - 300$ MeV is the confinement scale of QCD, the energy scale below below which quarks and gluons no longer are free.

It's appropriate here to mention about the running coupling constant ($\alpha_s(Q^2)$) in QCD, which depends on Q^2 .

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2) \beta_2 \ln(Q^2/\mu^2)} \quad (2.2.4)$$

where

$$\beta_2 = \frac{11N_c - 2N_f}{12\pi}$$

and μ is an arbitrary scale (renormalization scale). Above equation 2.2.4 can be rewritten

$$\alpha_s(Q^2) = \frac{1}{\beta_2 \ln \left(\frac{Q^2}{\Lambda_{\text{QCD}}^2} \right)} \quad (2.2.5)$$

with $N_c = 3$ the number of colors and N_f the number of quark flavors. The reason for using the variable N_f instead of the value 6 (from standard model) is that not all the quark flavors would participate based on the energy scale of the experiment. It can also be noted that as $Q^2 \rightarrow \infty$ the value of $\alpha_s(Q^2) \rightarrow 0$ implying the asymptotic freedom feature in QCD.

- **Bjorken variable (x_{Bj} or x):** It is the fraction of the light cone momentum of the proton carried by the struck quark. It is by definition bounded between $[0, 1]$. $x_{Bj} = 1$, implies that the photon sees a full proton (elastic regime), where as $x_{Bj} < 1$ implies that the photon probes part of the proton.

2.2.2 Cross-section and Structure functions

Recalling the relation between Cross-section and Matrix element from section 1.3, and using the covariant gauge for the photon propagator we can write the amplitude for the DIS process pictured in Fig. 2.2 as

$$i\mathcal{M}_{\sigma,\lambda,\lambda'}(X) = \frac{ie^2}{q^2} \bar{u}_{\lambda'}(p')\gamma^\mu u_\lambda(p) \langle X|J_\mu(0)|P,\sigma\rangle \quad (2.2.6)$$

Here λ and λ' are the electron polarizations before and after the interaction, and σ is the polarization of the proton. The initial state of the proton is denoted as $|P,\sigma\rangle$, while the final state of the many produced hadrons X in Fig 2.2 is correspondingly denoted as $|X\rangle$. As mentioned in the section 2.1, the protons wave function is treated as fock state as we will see moving forward.

We define the quark electromagnetic current by

$$J^\mu(x) = \sum_f Z_f \bar{q}_f(x)\gamma^\mu q_f(x) \quad (2.2.7)$$

where Z_f is the quark's electric charge in units of the electron charge e , $q_f(x)$ is the quark field operator, and the sum in the above equation runs over all quark flavors.

To calculate the total DIS cross section we need to square the amplitude, integrate or sum over the final-state quantum numbers, average over the initial-state quantum numbers, divide by the flux factor, and impose energy-momentum conservation, we get

$$\sigma^{ep} = \int \frac{d^3p'}{(2\pi)^3} \frac{1}{2E} \frac{1}{2E'} \frac{1}{4} \sum_{\sigma,\lambda,\lambda'} \sum_X |\mathcal{M}_{\sigma,\lambda,\lambda'}(X)|^2 (2\pi)^4 \delta^4(P + q - p_X) \quad (2.2.8)$$

Here P_X denotes the net four-momentum of all the hadrons produced in the scattering process.

Further calculations by substituting equation 2.2.6 in equation 2.2.8 gives us the expression for differential cross-section (in a compact form) as follows,

$$\frac{d\sigma}{d^3p'} = \frac{\alpha_{EM}^2}{EE'Q^4} L_{\mu\nu} W^{\mu\nu} \quad (2.2.9)$$

Where α_{EM} is the electromagnetic coupling constant, $L_{\mu\nu}$ is the leptonic tensor and $W^{\mu\nu}$ hadronic tensor.

The strong interaction dynamics in DIS (including nonperturbative contributions) is entirely contained in the hadronic tensor $W^{\mu\nu}$; therefore, it is very hard to calculate $W^{\mu\nu}$ in a “first principles” QCD calculation.

After some algebra, which involves conservation of electromagnetic currents one can show that the cross section of the reaction $e + p \rightarrow e' + X$ can be expressed in terms of the functions W_1 and W_2 .

$$\frac{d\sigma^{ep}}{dE' d\Omega} = \frac{\alpha_{EM}^2}{4E^2 \sin^4\left(\frac{\theta}{2}\right)} \left[W_2(x_{Bj}, Q^2) \cos^2\left(\frac{\theta}{2}\right) + 2W_1(x_{Bj}, Q^2) \sin^2\left(\frac{\theta}{2}\right) \right]. \quad (2.2.10)$$

where Ω is the solid scattering angle and θ is the electron scattering angle between $\bar{p}$ and $\bar{p}'$.

Here W_1 and W_2 are unknown scalar functions of x_{Bj} and Q^2 , called structure functions. The above equation demonstrates that the structure functions W_1 and W_2 can be measured experimentally by studying the angular dependence of the DIS cross section.

Note that the structure functions W_1 and W_2 have the dimension of inverse mass. It is more convenient to define dimensionless structure functions F_1 and F_2 , by

$$F_1(x_{Bj}, Q^2) \equiv mW_1(x_{Bj}, Q^2), \quad (2.2.11)$$

$$F_2(x_{Bj}, Q^2) \equiv \nu W_2(x_{Bj}, Q^2) = \frac{Q^2}{2mx_{Bj}} W_2(x_{Bj}, Q^2). \quad (2.2.12)$$

where

$$\nu \equiv \frac{P \cdot q}{m} = E - E'$$

2.2.3 Parton model and Bjorken scaling

To find the structure functions F_1 and F_2 it is easier to change the frame in which we are working. Instead of the proton's rest frame we will now use a frame in which the proton is ultra-relativistic. Such a frame is usually referred to as the infinite momentum frame (IMF) or Bjorken frame. One notable feature of the infinite momentum frame (IMF) is that, typical time scale of the quark and gluon (collectively known as partons and are assumed to be point-like) interactions inside the proton is much longer (due to time dilation) than the time scale of DIS. Hence, for the duration of the virtual photon-proton scattering, we can assume that the quarks and gluons do not interact with each other. Thus, the photon simply interacts with a quark in the proton.

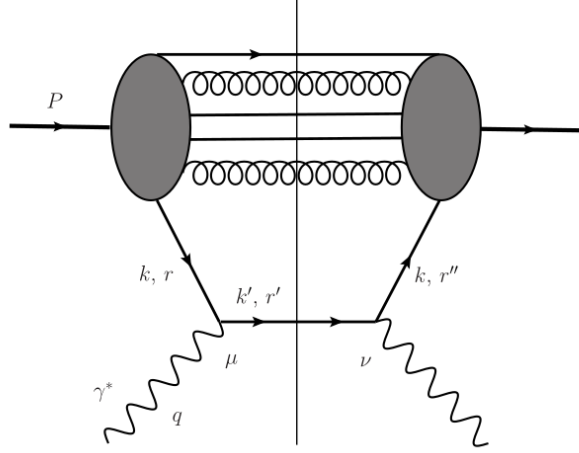


Figure 2.3: Virtual photon-proton scattering in the IMF

The full calculation of deep inelastic scattering (DIS) on a proton involves evaluating Feynman diagram in the infinite momentum frame (IMF) [2.3]. The labeling of the quark helicities r , r' , and r'' , k is the four momentum of the quark interacting with the photon. Below, I present the results that qualitatively serve the purpose of this thesis.

$$W_1(x_{Bj}) = \frac{1}{2m} \sum_f Z_f^2 q^f(x_{Bj}) \quad (2.2.13)$$

where $q^f(x_{Bj})$ is quark distribution function,

$$\begin{aligned} q^f(x_{Bj}) = & \frac{1}{2x_{Bj}} \sum_n \int d\xi d^2k_\perp \frac{1}{S_n} \sum_r \prod_{i=1}^n \frac{dx_i d^2k_{i\perp}}{x_i 2(2\pi)^3} \\ & \times \left| \Psi_n^f \left(\{x_i, k_{i\perp}\}; \frac{k^+}{P^+}, \vec{k}_\perp; r \right) \right|^2 \delta \left(1 - \xi - \sum_{l=1}^n x_l \right) \\ & \times \delta^{(2)} \left(\vec{k}_\perp + \sum_{j=1}^n \vec{k}_{j\perp} \right) \delta(x_{Bj} - \xi) \end{aligned} \quad (2.2.14)$$

the components of this function as one can see involve phase space integral, amplitude square of the fock state wave function (as mentioned in LCPT section), and delta functions providing insistance of conservation laws. also $\xi = \frac{k^+}{P^+}$ (origin of the definition of bjorken variable as given above in subsection 2.2.1).

The light-cone wave function of the $(n+1)$ -parton Fock state of the proton, denoted by $\psi_n^f(\{x_i, \vec{k}_{i\perp}\}; x, \vec{k}_\perp; r)$. The proton has n "spectator" partons (both quarks and gluons), and one quark carrying momentum

k in Fig 2.3 that interacts with the photon. This quark has helicity r and flavor f . In the expression 2.2.14 averaging over the proton polarizations and summation over the polarization, color, and flavor of the partons will always be implicitly assumed.

It turns out that $W_1(x_{Bj})$ and $W_2(x_{Bj})$ are connected in the following way for spin- $\frac{1}{2}$ quarks

$$\nu W_2(x_{Bj}) = 2mx_{Bj}W_1(x_{Bj}) \quad (2.2.15)$$

Also, from equations 2.2.11,

$$F_2(x_{Bj}) = 2x_{Bj}F_1(x_{Bj}) \quad (2.2.16)$$

The above equation is known as Callan–Gross relation. This relation would be different if the proton had constituents with a different spin interacting with the virtual photon.

One thing to note is that the structure functions depend only on the bjorken variable (x_{Bj}). This is a consequence of the important assumption of the parton model (naive) which is that the integrals in equation 2.2.14 are convergent even if we impose no integration limit on the transverse momentum integrals. And this is the only source for Q^2 term to pop in. This characteristic of the structure functions F_1 and F_2 turn out to depend only on one variable, x_{Bj} is called Bjorken scaling. Experimental results do support this theoretical prediction.

Final expressions for the structure functions are,

$$F_1(x_{Bj}) = \frac{1}{2} \sum_f Z_f^2 q^f(x_{Bj}), \quad (2.2.17)$$

$$F_2(x_{Bj}) = \sum_f Z_f^2 x_{Bj} q^f(x_{Bj}). \quad (2.2.18)$$

The functions F_1 and F_2 have a very simple physical meaning: namely, F_1 gives the number of partons in the hadron with longitudinal momentum fraction x_{Bj} (weighted by $Z_f^2/2$), while F_2 gives the average longitudinal momentum fraction of the partons in the hadron (weighted by Z_f^2) times the number of partons.

2.3 QCD evolution equations

Let us first rewrite the quark distribution function for a quark of flavor f from equation 2.2.14 as,

$$\begin{aligned}
q^f(x, Q^2) &= \sum_n \frac{1}{x} \int \frac{d^2 k_\perp}{2(2\pi)^3} \frac{1}{S_n} \sum_{\sigma=\pm 1} \prod_{i=1}^n \frac{dx_i}{x_i} \frac{d^2 k_{i\perp}}{2(2\pi)^3} \\
&\times |\Psi_n^f(\{x_i, k_{i\perp}\}; x, k_\perp; \sigma)|^2 (2\pi)^3 \\
&\times \delta^{(2)}\left(\vec{k}_\perp + \sum_{j=1}^n \vec{k}_{j\perp}\right) \delta\left(1 - x - \sum_{l=1}^n x_l\right). \tag{2.3.1}
\end{aligned}$$

Analogously, we can define the gluon distribution function as,

$$\begin{aligned}
G(x, Q^2) &= \sum_n \frac{1}{x} \int \frac{d^2 k_\perp}{2(2\pi)^3} \frac{1}{S_n} \sum_{\sigma=\pm 1} \prod_{i=1}^n \frac{dx_i}{x_i} \frac{d^2 k_{i\perp}}{2(2\pi)^3} \\
&\times |\Psi_n^f(\{x_i, k_{i\perp}\}; x, k_\perp; \lambda)|^2 (2\pi)^3 \\
&\times \delta^{(2)}\left(\vec{k}_\perp + \sum_{j=1}^n \vec{k}_{j\perp}\right) \delta\left(1 - x - \sum_{l=1}^n x_l\right). \tag{2.3.2}
\end{aligned}$$

Here, $\Psi_n(\{x_i, \vec{k}_{i\perp}\}; x, \vec{k}_\perp; \lambda)$ is the proton light-cone wave function containing n “spectators” along with a measured gluon having longitudinal momentum fraction x , transverse momentum $\vec{k}_\perp$, and polarization λ . S_n is the symmetry factor which involves inverse of factorials accounting for number of permutations of identical partons as since we are summing over we are over counting states which are not distinct.

The above quark and gluon distribution functions are collectively known as Parton Distribution Functions (PDF’s) which entails the information of probability of finding a parton.

This section presents evolution of parton distribution functions with respect to the key kinematic variables x and Q^2 . These are essentially differential equations which, when solved with initial conditions (typically obtained from experiments), describe the evolution of the distributions at other values of kinematic variables.

2.3.1 The Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) evolution equation

Let us now see what happens when the assumption mentioned above is lifted, which makes the quark distribution functions depend on Q^2 also. This causes violation of Bjorken scaling. Unlike in the naive parton model (it is assumed that the $k_\perp$ -integral is convergent in the UV, owing to some, presumed non-perturbative universal cutoff) the quark distribution function now depends on the momentum scale Q^2

(we have to cut off the $k_\perp$ -integral at Q^2 in the ultraviolet (UV), this constraint on transverse momentum appears due to dispersion relation), which enters equation 2.3.1 and Eq 2.3.2 as the renormalization scale. In the naive parton model, we assumed that the $k_\perp$ -integrals were sufficiently convergent that one could simply replace Q in the upper limit of integration by infinity without changing the value of the integral; this is, strictly speaking, only true for super-renormalizable theories and so is not true for QCD.

A series of calculations for the lowest-order corrections to the $\left| \Psi_n(\{x_i, \vec{k}_{i\perp}\}; x, \vec{k}_\perp; \sigma) \right|^2$ that involves solving all the Feynman diagrams (both real and virtual) for the four splitting reactions (mentioned below) are done and we end up with the DGLAP equations. The explicit calculations are beyond the scope of this thesis, but an idea of the process is given below.

The possible splitting reactions are as follows (Lowest order ones),

$$q \rightarrow q + g$$

$$q \rightarrow g + q$$

$$g \rightarrow q + \bar{q}$$

$$g \rightarrow g + g$$

where q : quark, $\bar{q}$: anti-quark and g : gluon.

Let us consider one of the possible Feynman diagrams for the first reaction ($q \rightarrow q + g$), where the produced quark (tagged) interacts with the virtual photon,

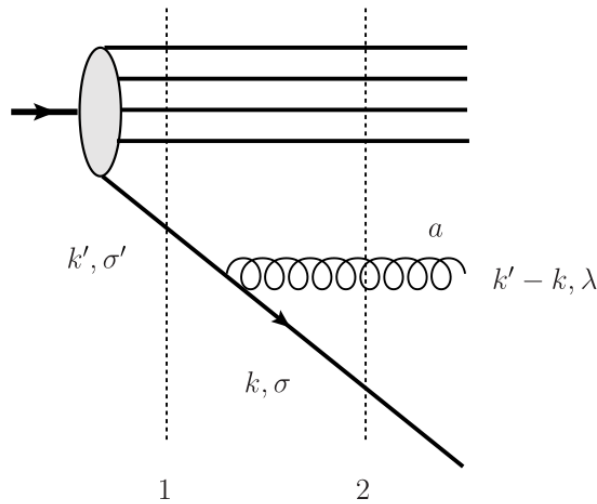


Figure 2.4: Lowest-order correction to the proton light cone wave function contributing to the quark distribution. The vertical dotted lines denote intermediate states.

Simple way to conclude and understand the calculations is that before the intermediate stage 1 (1st

dotted line) the wave function of the proton is $\Psi_{n-1}^f(\{k_i^+, \vec{k}_{i\perp}\}; x', \vec{k}'_\perp; \sigma')$ where $n-1$ are spectator partons and one is the parton which splits as per the diagram and the wave function before intermediate stage 2 (2nd dotted line) is $\Psi_n^f(\{k_i^+, \vec{k}_{i\perp}\}; x, \vec{k}_\perp; \sigma)$ where n are spectator partons and the tagged quark is the one interacting with photon. k and k' are four momentas and σ and σ' are polarizations.

It turns out the relation between both the above wave functions is as follows,

$$\Psi_n^f(\{k_i^+, \vec{k}_{i\perp}\}; x, \vec{k}_\perp; \sigma) \propto g \Psi_{n-1}^f(\{k_i^+, \vec{k}_{i\perp}\}; x', \vec{k}'_\perp; \sigma') \quad (2.3.3)$$

g is the QCD coupling and $\alpha_s = \frac{g_s^2}{4\pi}$.

As the quark distribution function has modulus square of the wave function this generates g^2 bringing in the factor of α_s . Also there is the $k_\perp$ integral that is of the form $\int_{\Lambda_{QCD}^2}^{Q^2} \frac{dk_\perp^2}{k_\perp^2}$ which gives a factor of $\ln\left(\frac{Q^2}{\Lambda_{QCD}^2}\right)$. The combined factor $\alpha_s \ln\left(\frac{Q^2}{\Lambda_{QCD}^2}\right) \sim 1$ which is the resummation of the parameter and is called the leading logarithmic approximation (LLA).

Note that the above calculation corresponds to a single Feynman diagram. The full result, obtained by summing contributions from all relevant Feynman diagrams, leads to a differential equation for the quark distribution function as follows:

$$Q^2 \frac{\partial q^f(x, Q^2)}{\partial Q^2} = \frac{\alpha_s C_F}{2\pi} \left[\int_x^1 \frac{dz}{z} \frac{1+z^2}{1-z} q^f\left(\frac{x}{z}, Q^2\right) - \int_0^1 dz \frac{1+z^2}{1-z} q^f(x, Q^2) \right]. \quad (2.3.4)$$

where

$$C_F = \frac{N_c^2 - 1}{2N_c}$$

is the Casimir operator in the fundamental representation of $SU(N_c)$. and $z = \frac{k^+}{k'^+}$.

$$P_{qq}(z) \equiv C_F \left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right]$$

is defined as quark-quark splitting function, with the “plus” notation defined as

$$\int_x^1 dz \frac{1}{(1-z)_+} f(z) = \int_x^1 dz \frac{1}{1-z} [f(z) - f(1)] + f(1) \ln(1-x), \quad (2.3.5)$$

for an arbitrary function $f(z)$ defined for $0 \leq x \leq 1$. With the help of $P_{qq}(z)$, equation 2.3.4 in the more compact form

$$Q^2 \frac{\partial q^f(x, Q^2)}{\partial Q^2} = \frac{\alpha_s}{2\pi} \int_x^1 \frac{dz}{z} P_{qq}(z) q^f\left(\frac{x}{z}, Q^2\right). \quad (2.3.6)$$

This is a differential equation for the quark distribution function. The initial condition for this equation is usually given by the quark distribution $q^f(x, Q_0^2)$ at some initial virtuality Q_0^2 . At low Q_0^2 , such an initial condition is likely to be due to some non-perturbative (large- α_s) physics: it cannot be calculated using perturbative techniques and is usually inferred from the data. Given the initial condition $q^f(x, Q_0^2)$, equation 2.3.6 allows one to uniquely construct the quark distribution function at all $Q^2 > Q_0^2$ (with leading-logarithmic accuracy). Therefore, equation 2.3.6 evolves the quark distribution function in Q^2 from some initial value at Q_0^2 to its value at another scale Q^2 : equations like 2.3.6 are usually referred to as **evolution equations**. The variation of a distribution function with Q^2 is known as the Q^2 -*evolution* of the distribution function.

Now one can see that the splitting processes (including the other three) should also contribute to the gluon distribution function. Therefore, to derive the evolution equations, we must account for all contributions from every relevant splitting process. Accounting for all these the final set of DGLAP equations are as follows:

$$Q^2 \frac{\partial \Delta^{f\bar{f}}(x, Q^2)}{\partial Q^2} = \frac{\alpha_s(Q^2)}{2\pi} \int_x^1 \frac{dz}{z} P_{qq}(z) \Delta^{f\bar{f}}\left(\frac{x}{z}, Q^2\right). \quad (2.3.7)$$

$$Q^2 \frac{\partial}{\partial Q^2} \begin{pmatrix} \Sigma(x, Q^2) \\ G(x, Q^2) \end{pmatrix} = \frac{\alpha_s(Q^2)}{2\pi} \int_x^1 \frac{dz}{z} \begin{pmatrix} P_{qq}(z) & P_{qG}(z) \\ P_{Gq}(z) & P_{GG}(z) \end{pmatrix} \begin{pmatrix} \Sigma(x/z, Q^2) \\ G(x/z, Q^2) \end{pmatrix}. \quad (2.3.8)$$

where

$$\Delta^{f\bar{f}}(x, Q^2) = q^f(x, Q^2) - q^{\bar{f}}(x, Q^2) \quad (2.3.9)$$

is flavor non-singlet distribution function, and

$$\Sigma(x, Q^2) = \sum_f \left[q^f(x, Q^2) + q^{\bar{f}}(x, Q^2) \right]. \quad (2.3.10)$$

is flavor singlet distribution function.

The other splitting functions are as follows,

$$P_{Gq}(z) = C_F \frac{1 + (1 - z)^2}{z}, \quad (2.3.11)$$

$$P_{qG}(z) = N_f [z^2 + (1 - z)^2], \quad (2.3.12)$$

$$P_{GG}(z) = 2N_c \left[\frac{z}{(1 - z)_+} + \frac{1 - z}{z} + z(1 - z) \right] + \frac{11N_c - 2N_f}{6} \delta(1 - z). \quad (2.3.13)$$

where $\bar{f}$ denotes the antiquark of flavor f . The flavor singlet and non-singlet functions are cleverly constructed to accommodate quark and anti-quark distributions, for example, the splitting of a gluon into $q\bar{q}$ pairs contributes equally to the creation of quarks and anti-quarks in the proton's wave function, it should not contribute to the non-singlet distribution $\Delta^{f\bar{f}}(x, Q^2)$.

P_{ba} means splitting function for the reaction $a \rightarrow b$ and the definition is, it is proportional to the probability of finding b in a 's wave function where $a, b \in [q, G]$.

Above DGLAP equations are valid only at the leading-logarithmic level. They are often referred to as the leading-order (LO) DGLAP equations, since the integral kernel on the right-hand side is given at the lowest order in α_s (i.e., at order α_s). Higher-order corrections to the splitting functions would generate terms with higher powers of α_s . Calculations up to $\mathcal{O}(\alpha_s^2)$, they are referred to as next-to-leading-order DGLAP or simply NLO DGLAP. The next order after that is called next-to-next-to-leading-order DGLAP (NNLO DGLAP), etc.

2.3.2 Double logarithmic approximation

Our whole effort of going through the idea of DGLAP evolution equations is for the below insight which we can infer at small- x .

Our main focus is QCD at high energy, i.e at high $\hat{s}$, which corresponds to small x (small- x physics). At this limit, small x implies small z regime, it turns out that only $P_{Gq}(z)$ and $P_{GG}(z)$ splitting functions are singular.

$$P_{Gq}(z)|_{z \ll 1} \approx \frac{2C_F}{z}, \quad (2.3.14)$$

$$P_{GG}(z)|_{z \ll 1} \approx \frac{2N_c}{z}. \quad (2.3.15)$$

This makes the evolution of the gluon distribution $G(x, Q^2)$ runs much faster than that of the quark distributions (both singlet and non-singlet), at small x . Therefore we can neglect the evolution of the quark distribution functions compared with that of the gluon. Also, the quark contribution to the gluon evolution, which enters via $P_{Gq}(z)$ into equation 2.3.8, is negligible as well: as is small owing to the lack of small- x enhancement to its own evolution, it would not contribute much to the gluon evolution.

Neglecting the quark distribution in the DGLAP equation 2.3.8 and using the approximation for the gluon-gluon splitting function from equation 2.3.14, we can write down an evolution equation for the gluon distribution only as,

$$Q^2 \frac{\partial G(x, Q^2)}{\partial Q^2} = \frac{\alpha_s(Q^2)}{2\pi} \int_x^1 \frac{dz}{z} \frac{2N_c}{z} G\left(\frac{x}{z}, Q^2\right) \quad (2.3.16)$$

which of course is valid only at small x .

Now changing variables, $z = \frac{k^+}{k'^+} = \frac{x}{x'}$ [since, $x = \frac{k^+}{P^+}$] and rewriting the above equation, we get

$$Q^2 \frac{\partial x G(x, Q^2)}{\partial Q^2} = \frac{\alpha_s(Q^2) N_c}{\pi} \int_x^1 \frac{dx'}{x'} x' G(x', Q^2). \quad (2.3.17)$$

Differentiating equation 2.3.17 with respect to $\ln(1/x)$, we can write it as

$$\frac{\partial x G(x, Q^2)}{\partial \ln(1/x) \partial \ln(Q^2/Q_0^2)} = \frac{\alpha_s(Q^2) N_c}{\pi} x G(x, Q^2) \quad (2.3.18)$$

with Q_0 a constant initial-virtuality scale.

For simplicity, let us imagine that the coupling constant is fixed, $\alpha_s(Q^2) = \alpha_s$. We can see from equation 2.3.18 that its solution iterates powers of α_s multiplied not just by one logarithm, $\ln(Q^2/Q_0^2)$, as in the DGLAP equations, but by two logarithms, $\ln(1/x) \ln(Q^2/Q_0^2)$. Thus the resummation parameter of equation 2.3.18 is

$$\alpha_s \ln \frac{1}{x} \ln \frac{Q^2}{Q_0^2}. \quad (2.3.19)$$

Thus at small coupling $\alpha_s \ll 1$, large $Q^2 \gg Q_0^2$, and small x such that $\ln(1/x) \gg 1$, we see that the small coupling α_s is multiplied by two large logarithms, which makes the resummation parameter 2.3.19 large and important to resum. The resummation of a series in powers of the parameter 2.3.19 is called the double logarithmic approximation (DLA).

The solution of equation 2.3.18 turns out to be

$$xG(x, Q^2) \sim \exp \left\{ 2 \sqrt{\frac{N_c}{\pi\beta_2} \ln \left(\frac{\ln(Q^2/\Lambda_{\text{QCD}}^2)}{\ln(Q_0^2/\Lambda_{\text{QCD}}^2)} \right) \ln \left(\frac{1}{x} \right)} \right\}. \quad (2.3.20)$$

The important inference from the above equation is that $xG(x, Q^2)$ increases as x decreases and/or Q^2 increases. The rise in $xG(x, Q^2)$ with decreasing x is therefore a prediction of the DGLAP evolution. This prediction agrees well with the experimental data at small- x as well.

2.3.3 Insight into other evolution equations

Note that the DGLAP evolution equations are with respect to Q^2 and now I will introduce few evolution equations that are with respect to the other kinematic variable, Bjorken $-x$, well more explicitly evolution with respect to Y (rapidity), where $Y = \ln(\frac{1}{x})$.

So an idea of how the evolution with respect to Y proceeds is from the double logarithmic approximation.

The small- x asymptotics of the gluon distribution function $xG(x, Q^2)$ in the framework of DGLAP evolution. For the LLA DGLAP, the small- x asymptotics corresponds to the summation of the parameter

$$\alpha_s \ln \left(\frac{1}{x} \right) \ln \left(\frac{Q^2}{Q_0^2} \right). \quad (2.3.21)$$

which constitutes the double logarithm approximation (DLA).

The asymptotics of the gluon distribution 2.3.20 is valid in the double logarithmic limit of small x and large Q^2 . However, if one is interested in studying the high energy (Regge) limit of QCD, one simply needs to fix Q^2 at some, not necessarily large, value and study the small- x asymptotics. As there is no need to take the large- Q^2 limit, $\ln(Q^2/Q_0^2)$ is now neither a large nor a small parameter. We therefore drop it from equation 2.3.21 and aim to resum the parameter

$$\frac{1}{\alpha_s} \ln \left(\frac{1}{x} \right) \quad (2.3.22)$$

The resummation of a series in powers of the parameter 2.3.22 is called the leading logarithmic approximation (LLA) in $1/x$. we assume that the relevant transverse momentum scales are large enough that $\alpha_s \ll 1$. At small x , we have $\ln(1/x) \gg 1$, so that $\alpha_s \ln(1/x) \sim 1$ and is an important parameter to resum. Since Q^2 is now fixed the parameter available to vary is the rapidity (Y) and hence the evolution equations which follow are evolutions over the rapidity (Y).

There are other evolution equations, namely the Balitsky–Fadin–Kuraev–Lipatov (BFKL) equation, the

nonlinear Gribov–Levin–Ryskin and Mueller–Qiu (GLR–MQ) equation, and the Balitsky–Kovchegov (BK) equation — all of which are developed for the small- x regime. The BK equation is developed using mueller’s dipole model. A detailed derivation of these equations is beyond the scope of this thesis and is not essential for the present project. However, a brief outline tracing the progression from one equation to another is provided in the appendix.

Chapter 3

Quantum Information Theory for DIS - An Overview

As mentioned in section 1.5, QIT tools are employed in understanding high energy interactions, in the hope of getting new insights into the long-standing problem of describing hadrons in terms of their fundamental constituents.

3.1 Theoretical Framework

The light-cone quantization emerged as a useful tool to describe hadrons in terms of their quark and gluon constituents. The hadron (proton) wave function on the light cone is given by,

$$|\Psi\rangle = \sum_{n=1}^{\infty} \int d\Gamma_n \Psi_n(x_i, \vec{k}_{\perp i}) \prod_{i=1}^n a_i^\dagger(x_i, \vec{k}_{\perp i}) |0\rangle, \quad (3.1.1)$$

we define Fock states with fixed number of partons n :

$$|n\rangle = \frac{1}{\sqrt{n!}} \prod_i^n a_i^\dagger |0\rangle, \quad (3.1.2)$$

$a_i^\dagger(x_i, \vec{k}_{\perp i})$ are the parton creation operators, and $\Psi_n(x_i, \vec{k}_{\perp i})$ are the light-cone wave functions of Fock states with n partons and $d\Gamma_n$ is the phase space differential:

$$d\Gamma_n = \delta\left(1 - \sum_i^n x_i\right) \delta^{(2)}\left(\sum_i^n \vec{k}_{\perp i}\right) dx_1 \dots dx_n d\vec{k}_{\perp 1} \dots d\vec{k}_{\perp n}, \quad (3.1.3)$$

The wave function of the hadron 3.1.1 can be written compactly using Fock states 3.1.2 as

$$|\psi\rangle = \sum_n \int d\Gamma_n \Psi_n(x_i, \vec{k}_{\perp i}) |n\rangle. \quad (3.1.4)$$

The Fock states with a fixed number of partons form a complete set of states with

$$\sum_n \int d\Gamma_n |n\rangle \langle n| = I. \quad (3.1.5)$$

It is implicit to assume the hadron wavefunction to be pure (since in quantum mechanics, a proton in isolation is a pure state, an eigen state of QCD Hamiltonian) and the corresponding density matrix for the hadron is

$$\hat{\rho} = |\Psi\rangle \langle \Psi| = \sum_{n,n'} \int d\Gamma_n d\Gamma_{n'} \Psi_n^*(x'_i, \vec{k}'_{\perp i}) \Psi_n(x_i, \vec{k}_{\perp i}) |n\rangle \langle n'|. \quad (3.1.6)$$

In any high-energy interaction, in our case the DIS, it is impossible for the probe to read off the information about the phases ϕ_n of the individual Fock states $|n\rangle$ with n partons. This is because the interaction time with the proton at high energies is in general much shorter than the characteristic time scale $t_n \sim 1/E_n$ with which the Fock states evolve, the light front captures an instant snapshot of the hadron wave function and cannot detect its time evolution. So since the phases are unobserved degrees of freedom, we can trace out the phases from the density matrix to obtain a reduced parton density matrix.

$$\hat{\rho}_{\text{parton}} = \text{Tr}_{\phi} \hat{\rho} \quad (3.1.7)$$

This has to be done as in light cone formalism which involves corresponding haar measure $U(1)$ (a mathematical tool), as follows

Let us assume that the hadron is probed by a light wave front described by $x^- = 0$. We therefore know that the positions z_i and times t_i of interaction of light with individual partons satisfy $t_i - z_i = x_i^- = 0$, but z_i and t_i individually cannot be determined in this measurement – therefore, the density matrix 3.1.6 has to be traced over x_i^+ , with the corresponding Haar measure ($\frac{dx^+}{2\pi}$). This leads to the factor

$$\int \frac{dx^+}{2\pi} e^{i(P_n^- - P_{n'}^-)x^+} = \delta(P_n^- - P_{n'}^-), \quad (3.1.8)$$

where $P_n^- = \sum_{i=1}^n p_i^-$. The constraint $P_n^- = P_{n'}^-$ is clearly satisfied for an arbitrary configuration of

partons for $n = n'$, and thus the Haar-scrambled parton density matrix is given by:

$$\hat{\rho}_{\text{parton}} = \text{Tr}_{x^+} |\Psi\rangle \langle \Psi| = \sum_n \int d\Gamma_n |\Psi_n(x_i, \vec{k}_{\perp i})|^2 |n\rangle \langle n|. \quad (3.1.9)$$

Therefore the density matrix measurable in high-energy interactions is diagonal in the number of partons. The parton probabilities

$$p_n = \int d\Gamma_n |\Psi_n(x_i, \vec{k}_{\perp i})|^2. \quad (3.1.10)$$

So equation 3.1.9 can be written as

$$\hat{\rho}_{\text{parton}} = \sum_n p_n |n\rangle \langle n|. \quad (3.1.11)$$

Purity of the obtained reduced density matrix is

$$\gamma_{\text{parton}} = \text{Tr}(\rho_{\text{parton}}^2) = \sum_n p_n^2 < 1. \quad (3.1.12)$$

Hence, this parton model density matrix represents a and describes an information-scrambled, mixed state. Now calculating entropy for this gives a non zero value as opposed to the hadron pure state whose entropy by definition is zero.

$$S_E = - \sum_n p_n \ln p_n \quad (3.1.13)$$

This entropy is referred to as "**Entanglement Entropy**" and describes entanglement between the number of partons and the phase of the corresponding Fock component of the wave function. The tracing over of phases causes quantum information scrambling and the emergence of the entanglement entropy.

Quantum theory offers us tools to calculate expectation values of observables $\hat{\mathcal{O}}$ according to

$$\langle \hat{\mathcal{O}} \rangle = \text{Tr}(\hat{\mathcal{O}} \hat{\rho}_{\text{parton}}) = \sum_n p_n \langle n | \hat{\mathcal{O}} | n \rangle; \quad (3.1.14)$$

this recipe corresponds to the computation within the parton model. In particular, evaluating the expectation value of the parton number operator $\hat{\mathcal{O}} = \hat{n} = a^\dagger a$, we get the average number of partons

in a hadron (at a particular value of Bjorken x and scale Q^2 at which the parton probabilities p_n are evaluated)

$$\langle n \rangle = \text{Tr}(\hat{n} \hat{\rho}_{\text{parton}}) = \sum_n p_n n. \quad (3.1.15)$$

It is further identified that, $\langle n \rangle = xG(x, Q^2)$ for the gluon distribution (more on this in the next chapter)

3.2 Experimental Evaluation

Hadrons – the bound states of quarks and gluons – store information about the dynamics of strong interactions. An access to this information is provided by high energy collisions in which detectors of produced final states serve as information readout devices. The DIS process where charged hadrons are counted is called "Inclusive DIS". This readout process is necessarily accompanied by the degradation of information content, and thus the creation of Shannon entropy,

$$H = - \sum_n p_n \ln p_n \quad (3.2.1)$$

where p_n is the probability of a particular measurement outcome taken from the space of all possible outcomes, with $\sum_n p_n = 1$. H is the entropy of the final state produced in the DIS, an experimentally deduced quantity.

In general, the entanglement entropy may differ from the entropy of the final hadronic state. From several examples, it is hypothesized that the entanglement entropy (S_E) is equal to the entropy of the hadron state (H), and moreover that the hadron multiplicity distribution mirrors the parton occupation number distribution inside the proton. This is referred to as “local parton-hadron duality”.

This overview provides an understanding of entanglement entropy and the entropy of the final hadronic state. Calculating entanglement entropy requires assuming a theoretical model, deriving the corresponding probabilities, and comparing the resulting entropy with experimental data which serves to tell how well the model captures the underlying physics.

Chapter 4

Modeling QCD Evolution

Modeling is one of the key approaches to understanding the dynamics of a physical process. The basic idea behind modeling is to map a physical process onto an existing, solved, or well-understood model, calculate observables that are accessible within the framework of that model, and then compare them with experimental data. If the model's predictions match the experimental results, one hopes that it provides insight into the underlying dynamics of the physical process.

Mueller's dipole model, where in the rest frame of the proton (or nucleus), the virtual photon emitted by the electron in deep inelastic scattering (DIS) fluctuates into a quark-antiquark ($q\bar{q}$) pair—this pair acts as a color dipole. This dipole then scatters off the target (proton/nucleus). The separation between the quark and antiquark defines the transverse size of the dipole. Due to the large relative boost factor between virtual photon and proton, in the small- x limit, the transverse size of color dipoles is fixed during its interaction with the proton and the resulting S-matrix is diagonal in the transverse dipole size. Since the rest frame of proton is considered, there are only "valance" quarks and in the small- x regime gluon distribution is more. If the original $q\bar{q}$ dipole emits a gluon, this corresponds to splitting into two new dipoles. This dipole cascade forms the basis of Mueller's dipole model, where a parent dipole evolves into a tree of daughter dipoles. This cascade resums contributions proportional to $\alpha_s \ln(1/x)$, the leading logarithmic approximation (LLA) in $1/x$ 2.3.22. The statistical evolution of these dipoles is described by the BFKL equation a linear evolution equation for the dipole density that leads to exponential growth in gluon number with rapidity.

When viewed in the infinite momentum frame, the proton's wavefunction contains a sea of partons. As x decreases, BFKL evolution generates many "wee" gluons (low-momentum partons) via a cascade that originates from higher- x gluons. Each gluon is emitted from a previously existing one, forming a ladder-like structure in momentum space.

However, a major limitation of BFKL evolution is that it predicts an exponential increase in gluon density, which fails to explain the phenomenon of gluon saturation (due to gluon recombination) observed in experiments. To address this, the BK equation was developed which introduces a nonlinear correction to the BFKL framework. This nonlinear term represents gluon recombination processes, thereby incorporating saturation effects into the evolution. Below is a form of BK equation which we will work with.

$$\frac{d\sigma(Y)}{dY} = \Delta\sigma_{\text{BFKL}}(Y) - \Delta\sigma_{\text{nonlinear}}(Y) \quad (4.0.1)$$

where σ is the dipole scattering cross section at rapidity Y and Δ is the BFKL intercept (a parameter estimated numerically).

4.1 1+1 Toy model of non-linear QCD evolution

So here the dipole model is explored in 1+1 dimensions [6] (one space and one time dimension), one key assumption here is all the dipoles are of fixed size.

In not so small- x regime when the gluon distributions are not large enough, splitting process dominates the recombination in the evolution. Hence the non-linear term from the above equation 4.0.1 can be dropped and the equation simplifies to

$$\frac{d\sigma(Y)}{dY} = \Delta\sigma(Y) \quad (4.1.1)$$

where Δ is the BFKL intercept. This reproduces the power-like increase of the cross section with energy: $e^{\Delta Y} = (1/x)^\Delta$.

Let $P_n(Y)$ denote the probability to find n dipoles at rapidity Y . Then the recurrence relation is given as follows:

$$\frac{dP_n(Y)}{dY} = -\Delta n P_n(Y) + (n-1)\Delta P_{n-1}(Y) \quad (4.1.2)$$

This is a typical cascade equation in which the first term describes the depletion of the probability to find n dipoles due to the splitting into $(n+1)$ dipoles, while the second term describes the growth due to the splitting of $(n-1)$ dipoles into n dipoles.

$$dP_n(Y)/dY = -\Delta \left[\text{Diagram 1} \right] + \Delta \left[\text{Diagram 2} \right]$$

The diagram shows the evolution of the probability of n dipoles. The first term, $-\Delta P_n(Y)$, represents the loss of a dipole at rapidity Y . The second term, $\Delta P_{n-1}(Y)$, represents the gain of a dipole at rapidity $Y + dY$. Both diagrams show a chain of dipoles with a new dipole being added or removed at a specific rapidity.

Figure 4.1: The graphic form of the equation 4.1.2 for the probability of n dipoles occurring at rapidity Y

To analyze this, a generating function (common mathematical procedure) is defined:

$$Z(Y, u) = \sum_n P_n(Y) u^n \quad (4.1.3)$$

At the initial rapidity $Y = 0$, we have only one dipole, so $P_1(Y = 0) = 1$ and $\sum_{n>1} P_n(Y = 0) = 0$ (so the state is pure); at $u = 1$, $Z(Y, u = 1) = \sum_n P_n(Y) = 1$. These two properties determine the initial and boundary conditions for the generating function:

$$Z(Y = 0, u) = u, \quad Z(Y, u = 1) = 1 \quad (4.1.4)$$

equation (4.1.2) then becomes a partial differential equation:

$$\frac{\partial Z(Y, u)}{\partial Y} = -\Delta u(1 - u) \frac{\partial Z(Y, u)}{\partial u} \quad (4.1.5)$$

It is instructive to observe that equation 4.1.5 implies a nonlinear equation for $Z(Y, u)$. The general solution to equation 4.1.5 turns out to be of the form $Z(Y, u) = Z(u(Y))$; if we substitute this function into equation 4.1.5, the derivatives $\partial Z/\partial u$ on both sides cancel, and we obtain a differential equation for the function $u(Y)$. By using the first of the initial conditions in equation 4.1.4, we can then rewrite equation 4.1.5 for rapidity near the initial condition as:

$$\frac{\partial Z}{\partial Y} = -\Delta(Z - Z^2) \quad (4.1.6)$$

The solution to equation (4.1.5), given the initial conditions, is:

$$Z(Y, u) = \frac{ue^{-\Delta Y}}{1 + u(e^{-\Delta Y} - 1)} = ue^{-\Delta Y} \sum_{n=1}^{\infty} u^n (1 - e^{-\Delta Y})^n \quad (4.1.7)$$

Comparing coefficients, we find the probability distribution:

$$P_n(Y) = e^{-\Delta Y} (1 - e^{-\Delta Y})^{n-1} \quad (4.1.8)$$

The von Neumann entropy of the system using the Gibbs formula $S = -\sum_n P_n(Y) \ln P_n(Y)$, is calculated as follows

$$S = -\sum_n P_n(Y) [-\ln(e^{\Delta Y} - 1) + n \ln(1 - e^{-\Delta Y})] \quad (4.1.9)$$

Using the generating function, this can be expressed as:

$$S(Y) = \ln(e^{\Delta Y} - 1) \cdot Z(Y, u=1) + \ln\left(\frac{1}{1 - e^{-\Delta Y}}\right) \cdot u \frac{\partial Z(Y, u)}{\partial u} \Big|_{u=1} \quad (4.1.10)$$

Evaluating the above gives:

$$S(Y) = \ln(e^{\Delta Y} - 1) + e^{\Delta Y} \ln\left(\frac{1}{1 - e^{-\Delta Y}}\right) \quad (4.1.11)$$

At large rapidity, i.e., $\Delta Y \gg 1$, this simplifies to:

$$S(Y) \rightarrow \Delta Y \quad (4.1.12)$$

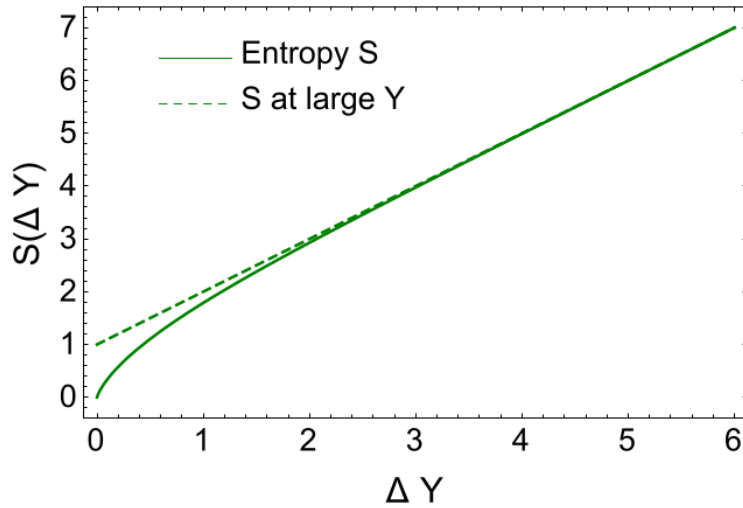


Figure 4.2: The entanglement entropy a function ΔY . The dashed line corresponds to the large rapidity limit of equation 4.1.11

The probability of finding n dipoles in this limit is

$$P_n(Y) \rightarrow e^{-\Delta Y} \quad (4.1.13)$$

The average number of partons (or gluons), which defines the parton distribution $xG(x)$, is:

$$xG(x) = \langle n \rangle = \sum_n n P_n(Y) = u \left. \frac{dZ(Y, u)}{du} \right|_{u=1} = e^{\Delta Y} = \left(\frac{1}{x} \right)^\Delta \quad (4.1.14)$$

Therefore, the entropy is simply related to the gluon distribution as:

$$S = \ln(xG(x)) \quad (4.1.15)$$

It is important to note that the relation in equation 4.1.15 emerges in the limit where all probabilities P_n become equal, see equation 4.1.13, which is due to the large rapidity limit. In this limit, an exponentially large number of partonic microstates occur with equal and exponentially small probabilities, $P_n(Y) = e^{-\Delta Y} = 1/\langle n \rangle$. It is well known that this equipartitioning of microstates maximizes the von Neumann entropy and corresponds to a maximally entangled state. We thus conclude that at small x , the proton represents a maximally entangled quantum state of partons.

Full calculation with all the steps : [Math-1.pdf](#)

This conclusion "that at small- x , the proton represents a maximally entangled quantum state of partons" is a novel result derived from theoretical calculations based on the underlying model. The next chapter generalizes this model and presents some experimental evaluations with few numerical results which estimates the model's accuracy.

4.2 3+1 dimensional Model

This section considers the cascade evolution in (3+1) dimensional QCD. The assumption of fixed dipole size in the 1+1 model is relaxed here. The cascade equation for the (3+1) case is

$$\begin{aligned} \frac{\partial P_n(Y; r_1, r_2, \dots, r_n)}{\partial \alpha_s \partial Y} = & - \sum_{i=1}^n \omega(r_i) P_n(Y; r_1, r_2, \dots, r_n) \\ & + \sum_{i=1}^{n-1} \frac{(r_i + r_n)^2}{(2\pi) r_i^2 r_n^2} P_{n-1}(Y; r_1, r_2, \dots, (r_i + r_n), \dots, r_{n-1}) \end{aligned} \quad (4.2.1)$$

where $P_n(Y; \{r_i\})$ is the probability to have n -dipoles with size r_i at rapidity $Y - y$. This QCD cascade leads to the Balitsky–Kovchegov equation for the amplitude and gives the theoretical description of the DIS.

By comparing equation 4.2.1 with equation 4.1.2 we can see that the probability for one dipole to survive is not a constant as in equation 4.1.2 but depends on the dipole size:

$$\bar{\alpha}_S \omega(r_i) \equiv \bar{\alpha}_S \omega_i = \frac{\bar{\alpha}_S}{2\pi} \int_{\rho} \frac{r_i^2}{(r_i - r')^2 r'^2} d^2 r' = \bar{\alpha}_S \ln \left(\frac{r_i^2}{\rho^2} \right), \quad (4.2.2)$$

where ρ is an infrared cutoff and $\bar{\alpha}_S = \alpha_S N_c / \pi$. The probability for a dipole of the size $|\vec{r}_1 + \vec{r}_2|$ to decay into two with the sizes r_1 and r_2 is equal to:

$$K(r_1, r_2 | r_1 + r_2) = \frac{\bar{\alpha}_S}{2\pi} \frac{(\vec{r}_1 + \vec{r}_2)^2}{r_1^2 r_2^2}. \quad (4.2.3)$$

The further calculation involves mathematical tools like Mellin Transform, feynman parameters etc and various assumptions, which are not necessary in our context as the next chapter focuses on the 1+1 model. But it turns out that the result for the entropy in this case is very similar to the one obtained above in the (1+1) toy model.

$$S \approx \bar{\alpha}_S z Y \quad (4.2.4)$$

which coincides with the expression 4.1.12 that we obtained for the toy model in the previous section if we take

$$\Delta = \bar{\alpha}_S z = \bar{\alpha}_S \ln(r^2 Q_s^2).$$

with $z \equiv \ln(r^2 Q_s^2)$. The characteristic dipole size r^2 in deep inelastic scattering is set by the momentum transfer Q^2 according to $r^2 \sim \frac{1}{Q^2}$.

Chapter 5

QCD evolution of Entanglement Entropy

In this chapter we shall see the 1+1 model, also referred to as Kharzeev-Levin (KL) model, put to use against two types of data sets. [7]

5.1 Entanglement Entropy in Inclusive DIS

Since we are employing 1+1 model, recall the expression for the probability to observe n dipoles at rapidity $Y = y$ (equation 4.1.8)

$$p_n(y) = \frac{e^{-\Delta y}}{C} \left(1 - \frac{e^{-\Delta y}}{C}\right)^{n-1}. \quad (5.1.1)$$

Here $y = \ln 1/x$ denotes the total available phase space $[0, y]$ in rapidity, i.e, the experiment starts at a particular Bjorken- x (corresponding rapidity y) and in the BFKL evolution, the kinematics are such that the longitudinal momenta are strongly ordered, while there are no constraints on the transverse momenta. This implies that particle emissions are strongly ordered in rapidity (decreasing order as dipoles are split), and therefore also in light-cone time. In contrast, DGLAP evolution involves strong ordering in transverse momenta, with no ordering in longitudinal momenta, essentially the reverse of the BFKL case.

Δ is the BFKL intercept of the $1 + 1$ dimensional dipole model. With $C = 1$, one fixes the initial conditions $p_1(0) = 1$, $p_{n>1}(0) = 0$. But the additional parameter C is introduced into equation 4.1.8, which can be adjusted to have a good fit with experimental data. As we will see one form of it C_0 , also

introduces non locality in the parton-hadron duality.

In the elastic limit $x = 1$ the system consists of a single dipole and one encounters a pure state. The mean number of dipoles is obtained as

$$\langle n \rangle_y = \sum_n n p_n(y) = C e^{\Delta y} \equiv \bar{n}(x). \quad (5.1.2)$$

When comparing to experimental data, it is important to take into account that only charged hadrons, predominantly charged pions, are observed in experiment. To compare to data, it is necessary to rescale the mean number of dipoles by a factor $\frac{2}{3}$ and use $C \rightarrow C' = \frac{2C}{3}$. Using color dipoles as a suitable basis for the Schmidt decomposition in the low- x DIS process, one finally obtains the following expression for the entanglement entropy in inclusive DIS:

$$S_{\text{inc.}}(\bar{n}) = - \sum_n p_n(\bar{n}) \ln p_n(\bar{n}) = \ln \bar{n} - (\bar{n} - 1) \ln \left(1 - \frac{1}{\bar{n}} \right) \quad (5.1.3)$$

The dipole probabilities are expressed in terms of average number of dipoles,

$$p_n(y) = p_n(\bar{n}(x)), \quad p_n(\bar{n}) = \frac{1}{\bar{n}} \left(1 - \frac{1}{\bar{n}} \right)^{n-1}. \quad (5.1.4)$$

from equation 5.1.4 it is evident that in the limit of large multiplicities $\bar{n} \gg 1$ the distribution turns into the homogeneous distribution which is characterized by maximal entanglement entropy $[S_{\text{inc.}}(\bar{n}) \sim \ln(\bar{n})]$.

It appears that the expression $S_{\text{inc.}}(\bar{n}) = \ln(\bar{n})$, from experimental calculation of hadron multiplicity distribution and identifying it with the average number of parton, is more general than merely the large $\bar{n}$ limit of equation 5.1.1. Hence

$$S_{\text{inc.}}^{\text{univ.}}(\bar{n}) = \ln(\bar{n}), \quad (5.1.5)$$

is not only a characteristic for this probability distribution equation 5.1.1, but also arises as the large $\bar{n}$ limit of other distributions, we refer in the following to it as the “universal entanglement entropy”.

The complete entropy of the KL model is from equation 5.1.3,

$$S_{\text{inc.}}^{KL}(\bar{n}) = \ln \bar{n} - (\bar{n} - 1) \ln \left(1 - \frac{1}{\bar{n}} \right) \quad (5.1.6)$$

As from equation 4.1.14 we identify the average number of particles as the total number of partons per $\ln(1/x)$, as obtained from the sum of quark and gluon parton distribution functions (PDFs):

$$\bar{n}(x, \mu) \equiv \frac{dn}{d \ln(1/x)} = x \Sigma(x, \mu) + x g(x), \quad (5.1.7)$$

$$\Sigma(x, \mu) = \sum_f (q_f(x, \mu) + \bar{q}_f(x, \mu)). \quad (5.1.8)$$

The μ dependence of PDFs is because of the running coupling constant α_s (2.2.4) and although at small x , gluon contribution is dominant the RHS includes quark flavor singlet function (sea quark) also as the experiment only determines joint combination.

The average number of dipoles is identified as,

$$\bar{n}(x, \mu) = \left(\frac{1}{x} \right)^{\Delta(\mu)}, \quad (5.1.9)$$

with the intercept Δ (another parameter besides C which was introduced earlier) obtained from a fit to parton distribution function in the low x region,

$$C \left(\frac{1}{x} \right)^{\Delta} = x \Sigma(x) + x g(x) \quad (5.1.10)$$

In both cases (universal and KL) the mean particle number will be rescaled by a factor 2/3 to take into account that observed hadrons are dominated by pions and only charged hadrons are observed in experiment. Numerically, both approaches are close.

The following subsections develops tools to describe a setup where dipoles are observed in a limited region of phase space. This experimentally achieved by moving rapidity window which captures the evolution of entanglement entropy.

5.1.1 Definition of a local dipole probability distribution

The probability distribution (5.1.4) with fixed dipole sized (KL model), with rapidity range $[0, y]$. We can divide this range into two intervals, $[0, y_0]$ and $[y_0, y]$, and count only emissions in the region $[y_0, y]$. To address this case, we introduce probabilities $\tilde{p}_n(y, y_0)$ as the probability to have n dipoles in the rapidity range $[y_0, y]$. With $p_n(y_0)$ being the probability to have n dipoles in the range $[0, y_0]$, these new probabilities are subject to the following consistency condition:

$$P_n(y) = \sum_{m=0}^n p_{n-m}(y_0) \tilde{p}_m(y, y_0) \quad (5.1.11)$$

The probability to have n dipoles in the interval $[0, y]$ is given by all possible combinations to have $n-m$ dipoles in the range $[0, y_0]$ and m in the range $[y_0, y]$. With

$$\langle n \rangle_{y; y_0} = \sum_n n \tilde{p}_n(y, y_0), \quad (5.1.12)$$

and $\sum_{m=0} \tilde{p}_m(y, y_0) = 1$, this yields immediately

$$\langle n \rangle_y = \langle n \rangle_{y_0} + \langle n \rangle_{y; y_0}. \quad (5.1.13)$$

The average number of dipoles emitted in the interval $[0, y]$ is given as the sum of the average numbers of dipoles emitted in the regions $[0, y_0]$ and $[y_0, y]$. The equation 5.1.11 can be solved by

$$\begin{aligned} \tilde{p}_0(y, y_0) &= e^{-\Delta(y-y_0)}, \\ \tilde{p}_{n \geq 1}(y, y_0) &= p_n(y) \cdot (1 - \tilde{p}_0(y, y_0)). \end{aligned} \quad (5.1.14)$$

Here $p_n(y)$ denote the inclusive dipole probabilities in equation 5.1.1, while $\tilde{p}_0$ yields the probability to have no emission into the interval $[y_0, y]$. Since $\tilde{p}_0(y, y_0)$ is the probability for no emission in the interval, $1 - \tilde{p}_0(y, y_0)$ is the probability to have any number of emissions in the interval. Comparing the above probabilities to the inclusive probabilities equation 5.1.4, it is further natural to express them as functions of the mean number of dipoles and the no-emission probability $\tilde{p}_0$:

$$\begin{aligned} \tilde{p}_{n \geq 1}(y, \tilde{p}_0) &= \tilde{p}_{n \geq 1}(\bar{n}(x), \tilde{p}_0), \\ \tilde{p}_{n \geq 1}(\bar{n}, \tilde{p}_0) &= p_n(y) \cdot (1 - \tilde{p}_0(y, y_0)). \end{aligned} \quad (5.1.15)$$

5.1.2 Definition of local entropy

Given the above distribution of dipoles in the region $[y_0, y]$, we can find the entropy of the particles emitted into the interval $[y_0, y]$ the following local entropy

$$S_{\text{loc}}(\bar{n}, \tilde{p}_0) = - \sum_{n=0} \tilde{p}_n(\bar{n}, \tilde{p}_0) \ln \tilde{p}_n(\bar{n}, \tilde{p}_0) \quad (5.1.16)$$

$$= -\tilde{p}_0 \ln \tilde{p}_0 - (1 - \tilde{p}_0) \ln(1 - \tilde{p}_0) \quad (5.1.17)$$

$$+ (1 - \tilde{p}_0) S_{\text{inc.}}(\bar{n}). \quad (5.1.18)$$

It is given as the sum of the entropy of a two-level system (emission into the rapidity interval or not) plus the entropy of the inclusive system times the probability to have emission into the interval. Note that the above expression does not require the particular form of the probabilities of the KL-model (the cascade equation). Indeed, as long as the probabilities of equation 5.1.4 satisfy $\sum_{n=1}^{\infty} p_n = 1$ together with the factorizing form equation 5.1.15, the equation 5.1.16 will hold. The above expression for the local entropy fulfills the following constraints: If $\tilde{p}_0 = 1$, the probability for emission into the interval $[y_0, y]$ is zero and consequently $S_{\text{loc}} \rightarrow 0$ in this limit. For $\tilde{p}_0 = 0$, the probability distribution for emission into the intervals When the intervals $[y_0, y]$ and $[0, y]$ coincide, one finds $S_{\text{loc}} = S_{\text{inc.}}$, as expected.

For the following numerical studies based on equation 5.1.16, it will be necessary to extend the expression for the no-emission probability $\tilde{p}_0$ to:

$$\tilde{p}_0 = C_0 e^{-\Delta(y-y_0)}, \quad (5.1.19)$$

with $C_0 \leq 1$ a constant factor that accounts for subleading effects that modify the no-emission probability.

With this parameter, in particular $\tilde{p}_0(y, y) = C_0 \neq 1$, i.e., the probability for emission $(1 - \tilde{p}_0)$ is non-zero even when the rapidity range $y - y_0$ shrinks to zero. This introduces a certain non-locality in the parton-hadron duality, which relates the generated dipoles to final-state hadrons. While the overall number of dipoles is fixed by the PDFs, a dipole in the range $[0, y_0]$ can generate, with a certain non-zero probability, a hadron in the range $[y_0, y]$.

Full calculation with all the steps : [Math-2.pdf](#)

5.2 Experimental data

As mentioned in the beginning of this chapter, there are two types of data, one is the moving rapidity window (in the laboratory frame), which is mentioned in the above subsection 5.1.1 . Another is the fixed detection window (in the Hadron center of mass frame) which corresponds to global data set kinematics

of the reaction. This is illustrated in the below figure 5.1

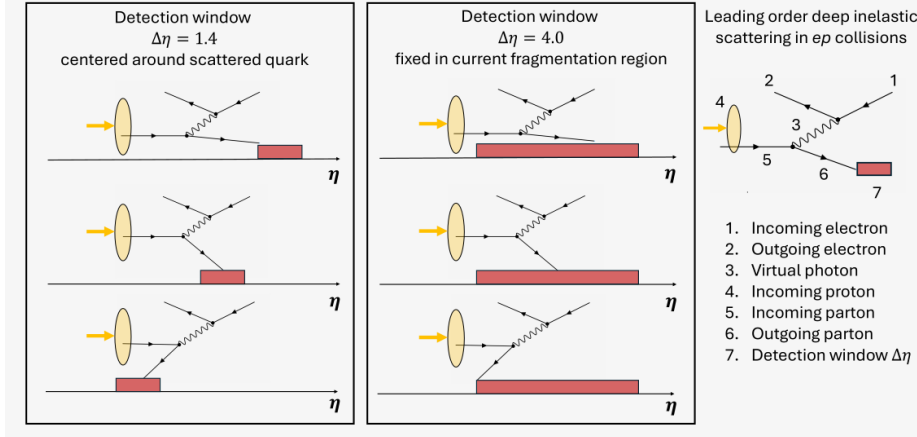


Figure 5.1: Illustration of the moving detection window (left) vs fixed detection window (right) in the H1 measurement

In the leading DIS picture, the leading quark can be estimated from the kinematic variables x and Q^2 , and the pseudorapidity ranges are corresponding to the respective x range.

Moving finite size rapidity window:

- Intuitively, the leading scattered quark goes more forward for high- x events (top left in Fig.5.1), as a high- x quark carries a larger momentum fraction of the incoming proton and consequently has a smaller scattering angle. The detection window of 1.4 units of pseudorapidity is therefore placed more forward. As the energy of the virtual photon increases at fixed virtuality Q^2 , the value of x decreases, and the scattering angle of the leading quark increases. Thus, the detection window moves backward, as illustrated in the next two sub-panels on the left in Fig. 5.1.

- A narrow detection window centered around the scattered quark with $\Delta\eta \approx y - y_0 = 1.4$.

Fixed detection window:

- For the fixed detection window, the position of the window is chosen to qualitatively correspond to the current fragmentation region. The current fragmentation region is then selected as the pseudorapidity range $0 < \eta^* < 4$, which must lie within the laboratory frame region $-1.6 < \eta_{\text{lab}} < 1.6$. Note that the lab-frame η selection corresponds to the boundary of the tracking acceptance of the H1 detector. Most of the current fragmentation region overlaps with this lab-frame η selection. This setup is illustrated in Fig. 5.1 on the right.

- A fixed detection window in the current fragmentation region with $\Delta\eta^* = 4.0$ ($\Delta\eta \approx 3.2$).
- The photon-going direction is defined as the positive pseudorapidity. η^* represents that the events

are viewed in hadronic center of mass (HCM) frame.

There are two sources for PDF's, one is Hadron Electron Ring Accelerator(HERA) PDF's and the other is (Hentschinski-Sabio Vera-Salas) HSS PDF's. The ep DIS data from the H1 detector, for both the types of detection window are presented in [8].

5.3 Numerical Results

Both the data sets are treated with their respective limitation of the range in which the data have been recorded i.e, $\Delta\eta \approx y - y_0 = 3.2$ (for fixed detection window) and $\Delta\eta \approx y - y_0 = 1.4$ (for moving detection window). This will test the generalization of the KL model made in above subsections 5.1.1 and 5.1.2.

Below is the procedure followed based on the data sets and equations from the KL model:

- The parameters C and the intercept Δ are determined (See Table 5.1) from a fit to leading order PDF's, using the relation in equation 5.1.10 in low- x limit. For the parton distributions (R.H.S of equation 5.1.10) leading order HERAPDFs [9] and PDFs obtained from the HSS unintegrated gluon distribution [10], [11] and [12] are used. For both PDF sets the hard scale of the photon virtuality is identified to be $Q = \mu$.
- The uncertainties on the extracted parameters are estimated by varying the hard scale μ in the interval $\mu \in [Q/2, 2Q]$. Further, the range in x over which the fits are performed is varied, i.e., the ranges $x \in [10^{-2}, 10^{-4}]$ and $x \in [10^{-4}, 10^{-6}]$ are taken. Uncertainties due to the choice of the hard scale and uncertainties due to a variation in the range of x are added in quadrature to yield the complete uncertainty.

$Q^2/\text{GeV}^2 \in$	[5, 10]	[10, 20]	[20, 40]	[40, 100]
Δ_{HSS}	0.27 ± 0.04	0.28 ± 0.04	0.29 ± 0.05	0.29 ± 0.04
Δ_{HERAPDF}	0.28 ± 0.08	0.31 ± 0.07	0.33 ± 0.06	0.36 ± 0.06
C_{HSS}	2.1 ± 0.3	2.7 ± 0.4	3.2 ± 0.5	3.9 ± 0.7
C_{HERAPDF}	2.54 ± 0.68	2.62 ± 0.78	2.65 ± 0.91	2.66 ± 1.02

Table 5.1: Parameters C and Δ extracted from a comparison to HSS and HERA PDFs.

- The above BFKL intercept Δ is extracted for each PDF, $\Delta\eta \approx y - y_0 = 3.2$ (for fixed window) and $\Delta\eta \approx y - y_0 = 1.4$ (for moving window). Substituting them in equation 5.1.19 gives $\bar{p}_0$ in terms of C_0 . From equation 5.1.16, $S_{inc}(\bar{n})$ can be evaluated once with $S_{inc}^{KL}(\bar{n})$ and once with $S_{inc}^{univ.}(\bar{n})$. Corresponding C_0 's are listed in table 5.2.

C_0	HSS PDFs	LO HERAPDF
KL model	0.655 ± 0.009	0.777 ± 0.008
universal	0.777 ± 0.008	0.706 ± 0.008

Table 5.2: Values of the constant C_0 obtained from a fit to data. The uncertainties are directly related to the uncertainties of the underlying data set.

The value of C_0 is roughly 0.7 i.e, the emission probability at $y = y_0$ is of the order of 0.3, which is in accordance with the expected value of the strong coupling constant α_s .

Hadronic entropy as obtained from equation 5.1.16 employing equation 5.1.5 (moving rapidity window) and for the inclusive entropy (fixed rapidity window) with the number of dipoles determined through equation 5.1.7 based on leading order HERAPDF's is shown in below figure 5.2 .

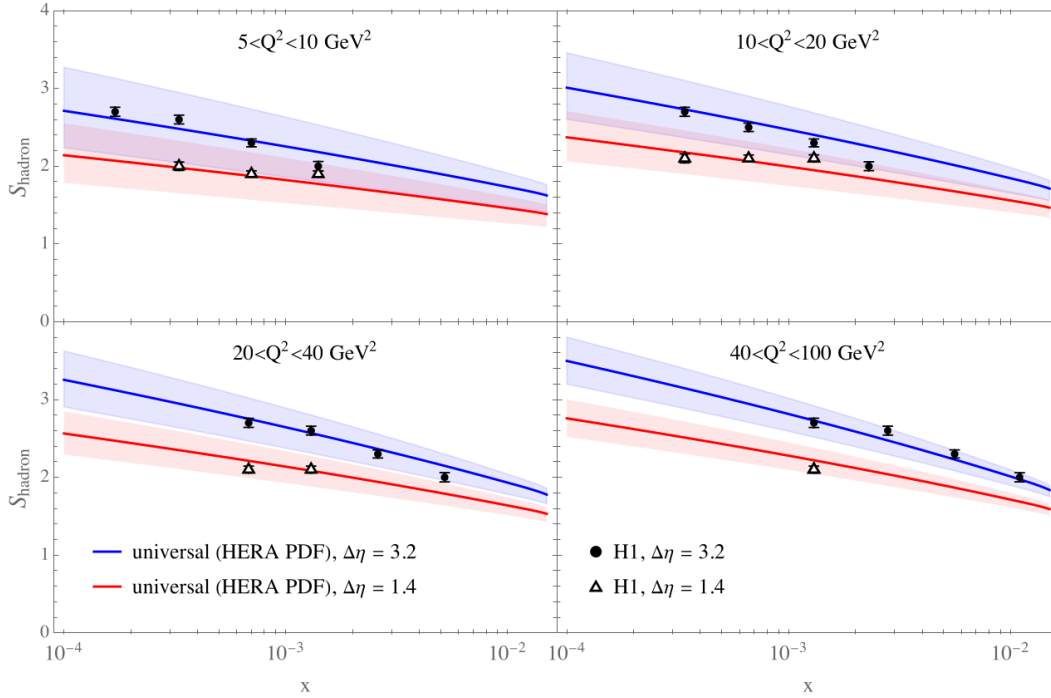


Figure 5.2: Experimental data and theoretical predictions

The points represent the hadron entropy calculated from experiments (3.2) for both the data sets obtained from H1 detector, for various ranges of Q^2 . The lines in the figure 5.2 represent the evolution of universal entropy (according to the model). This plot shows the comparison of the theoretical predictions to experimental data based on HERAPDF using the universal entropy formula 5.1.5 .

χ^2 (a statistical quantity which gives the accuracy of the fit) tables for the fits are also presented [7]

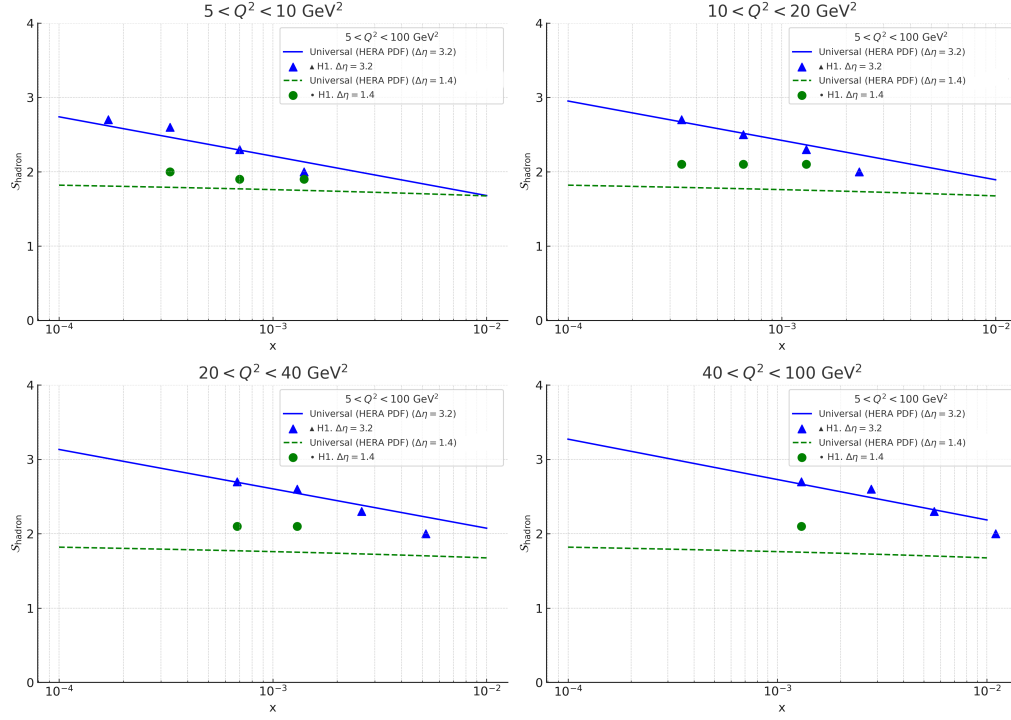


Figure 5.3: Replicated version of the plot shown in Figure 5.2.

Below are the parameters and procedure used to replicate the plots. The parameters are well within the bounds specified in Tables 5.1 and 5.2 . S_{hadron} data is taken from [8].

Q^2 Range (GeV^2)	$(\Delta_{\text{HERA PDF}}, C_{\text{HERA PDF}})$ at $\Delta\eta = 3.2$	$(\Delta_{\text{HERA PDF}}, C_{\text{HERA PDF}}, C_0)$ at $\Delta\eta = 1.4$
$5 < Q^2 < 10$	(0.23, 1.86)	(0.36, 3.22, 0.698)
$10 < Q^2 < 20$	(0.23, 2.3)	(0.36, 3.22, 0.698)
$20 < Q^2 < 40$	(0.23, 2.76)	(0.36, 3.22, 0.698)
$40 < Q^2 < 100$	(0.236, 3.0)	(0.36, 3.22, 0.698)

Table 5.3: Fit parameters for various Q^2 ranges at fixed $\Delta\eta$ values

Using equations 5.1.5 and $\bar{n} = C(\frac{1}{x})^\Delta$ with above specified values of Δ and C corresponding to $\Delta\eta = 3.2$.

Using equations 5.1.5, 5.1.16 and 5.1.19 with above specified values of Δ , C and C_0 corresponding to $\Delta\eta = 1.4$. **Remark:** Best fits for the available ranges of parameters.

Although the replication of the plots are accurate for the case $\Delta\eta = 3.2$, a deviation is observed for $\Delta\eta = 1.4$, likely due to the direct substitution of $\Delta\eta = y - y_0$ in equation 5.1.19.

The code can be found in this [GitHub Repository](#).

Chapter 6

Closing Remarks

6.1 Conclusions

- The plot shows that both the local ($\Delta\eta = 1.4$) and inclusive ($\Delta\eta = 3.2$) cases are very well described by the KL model. It can therefore be argued that the generalized formalism can indeed account for the emission into a local rapidity window, and describe both the data taken for a moving and fixed rapidity window.
- It turns out that the model is also able to describe other QCD process, like diffractive DIS, and shows good agreement with the revised model of entanglement entropy [13]. This serves as an important confirmation of the model's ability to describe two different QCD processes within a consistent framework.
- It was remarked in section 4.1 that the model at small- x describes an exponentially large number of partonic microstates with equal and exponentially small probabilities, $P_n(Y) = e^{-\Delta Y} = 1/\langle n \rangle$. It is well known that such equipartitioning of microstates maximizes the von Neumann entropy and corresponds to a maximally entangled state. Since the model is able to explain wide range of data, this remark can be substantiated.

6.2 Future work

- To simulate the numerical results from the H1 data and replicate the above results.
- To explore if there are more accurate evolution equations or models which can fit varied experimental data.

Appendix

This appendix presents the expressions of Balitsky–Fadin–Kuraev–Lipatov (BFKL) equation, the non-linear Gribov–Levin–Ryskin and Mueller–Qiu (GLR-MQ) equation, and the Balitsky–Kovchegov (BK) equation without any explicit calculation. This ought to act as a bridge in understanding as to how the BK equation in chapter 4 has come about.

The BFKL equation:

This equation governs the evolution of the gluon Green’s function $G(\vec{l}_\perp, \vec{l}'_\perp, Y)$ with rapidity Y :

$$\frac{\partial G(\vec{l}_\perp, \vec{l}'_\perp, Y)}{\partial Y} = \frac{\alpha_s N_c}{\pi^2} \int d^2 q_\perp \frac{1}{(\vec{l}_\perp - \vec{q}_\perp)^2} \left[G(\vec{q}_\perp, \vec{l}'_\perp, Y) - \frac{l_\perp^2}{2q_\perp^2} G(\vec{l}_\perp, \vec{l}'_\perp, Y) \right] \quad (\text{A.1})$$

The non-linear GLR-MQ equation:

This equation includes nonlinear recombination effects to address unitarity violations in BFKL:

$$\frac{\partial^2 xG(x, Q^2)}{\partial \ln(1/x) \partial \ln(Q^2/\Lambda^2)} = \frac{\alpha_s N_c}{\pi} xG(x, Q^2) - \frac{\alpha_s^2 N_c \pi}{2C_F S_\perp Q^2} [xG(x, Q^2)]^2 \quad (\text{A.2})$$

- $S_\perp$: Transverse area of the proton or nucleus.
- The first term is linear (BFKL-like), and the second term is nonlinear due to gluon recombination (fusion).

The BK equation:

$$\frac{\partial N(\mathbf{x}_{01}, Y)}{\partial Y} = \frac{\alpha_s N_c}{2\pi^2} \int d^2 \mathbf{x}_2 \frac{x_{01}^2}{x_{02}^2 x_{12}^2} [N(\mathbf{x}_{02}, Y) + N(\mathbf{x}_{12}, Y) - N(\mathbf{x}_{01}, Y) - N(\mathbf{x}_{02}, Y)N(\mathbf{x}_{12}, Y)] \quad (\text{A.3})$$

- $N(\mathbf{x}_{ij}, Y)$: Dipole scattering amplitude for a dipole of transverse size $\mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j$ at rapidity Y .
- α_s : Strong coupling constant.
- N_c : Number of colors.
- $\mathbf{x}_2$: Transverse coordinate of the emitted gluon.
- $x_{ij}^2 = (\mathbf{x}_i - \mathbf{x}_j)^2$: Squared transverse distance between two points.

References

- [1] Donald H. Perkins. *Introduction to High Energy Physics*. 4th ed. Cambridge University Press, 2000. ISBN: 9780521621960.
- [2] Richard P. Feynman. *Photon-Hadron Interactions*. Reading, Massachusetts: Benjamin, 1972. ISBN: 9780805301427.
- [3] Yuri V. Kovchegov and Eugene Levin. *Quantum Chromodynamics at High Energy*. Vol. 33. Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology. Cambridge: Cambridge University Press, 2012. ISBN: 9780521112574. DOI: 10.1017/CB09781139022187.
- [4] Dmitri E. Kharzeev. “Quantum information approach to high energy interactions”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 380.2216 (Dec. 2021). ISSN: 1471-2962. DOI: 10.1098/rsta.2021.0063. URL: <http://dx.doi.org/10.1098/rsta.2021.0063>.
- [5] P. A. M. Dirac. “Forms of Relativistic Dynamics”. In: *Reviews of Modern Physics* 21.3 (1949), pp. 392–399. DOI: 10.1103/RevModPhys.21.392.
- [6] Dmitri E. Kharzeev and Eugene M. Levin. “Deep inelastic scattering as a probe of entanglement”. In: *Phys. Rev. D* 95 (11 June 2017), p. 114008. DOI: 10.1103/PhysRevD.95.114008. URL: <https://link.aps.org/doi/10.1103/PhysRevD.95.114008>.
- [7] Martin Hentschinski et al. “QCD evolution of entanglement entropy”. In: *Reports on Progress in Physics* 87.12 (Dec. 2024), p. 120501. ISSN: 1361-6633. DOI: 10.1088/1361-6633/ad910b. URL: <http://dx.doi.org/10.1088/1361-6633/ad910b>.
- [8] H1 collaboration. *Measurement of charged particle multiplicity distributions in DIS at HERA and its implication to entanglement entropy of partons*. 2020. arXiv: 2011.01812 [hep-ex]. URL: <https://arxiv.org/abs/2011.01812>.
- [9] H1 and ZEUS Collaborations. *Combination of Measurements of Inclusive Deep Inelastic $e^\pm p$ Scattering Cross Sections and QCD Analysis of HERA Data*. 2015. arXiv: 1506.06042 [hep-ex]. URL: <https://arxiv.org/abs/1506.06042>.

- [10] Martin Hentschinski, Agustín Sabio Vera, and Clara Salas. “Hard to Soft Pomeron Transition in Small- x Deep Inelastic Scattering Data Using Optimal Renormalization”. In: *Phys. Rev. Lett.* 110 (4 Jan. 2013), p. 041601. DOI: 10.1103/PhysRevLett.110.041601. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.110.041601>.
- [11] Martin Hentschinski, Agustín Sabio Vera, and Clara Salas. “ F_2 and F_L at small x using a collinearly improved BFKL resummation”. In: *Phys. Rev. D* 87 (7 Apr. 2013), p. 076005. DOI: 10.1103/PhysRevD.87.076005. URL: <https://link.aps.org/doi/10.1103/PhysRevD.87.076005>.
- [12] I. Bautista, A. Fernandez Tellez, and M. Hentschinski. “BFKL evolution and the growth with energy of exclusive J/Ψ and Υ photoproduction cross sections”. In: *Phys. Rev. D* 94 (5 Sept. 2016), p. 054002. DOI: 10.1103/PhysRevD.94.054002. URL: <https://link.aps.org/doi/10.1103/PhysRevD.94.054002>.
- [13] Martin Hentschinski et al. *Probing the onset of maximal entanglement inside the proton in diffractive DIS*. 2024. arXiv: 2305.03069 [hep-ph]. URL: <https://arxiv.org/abs/2305.03069>.